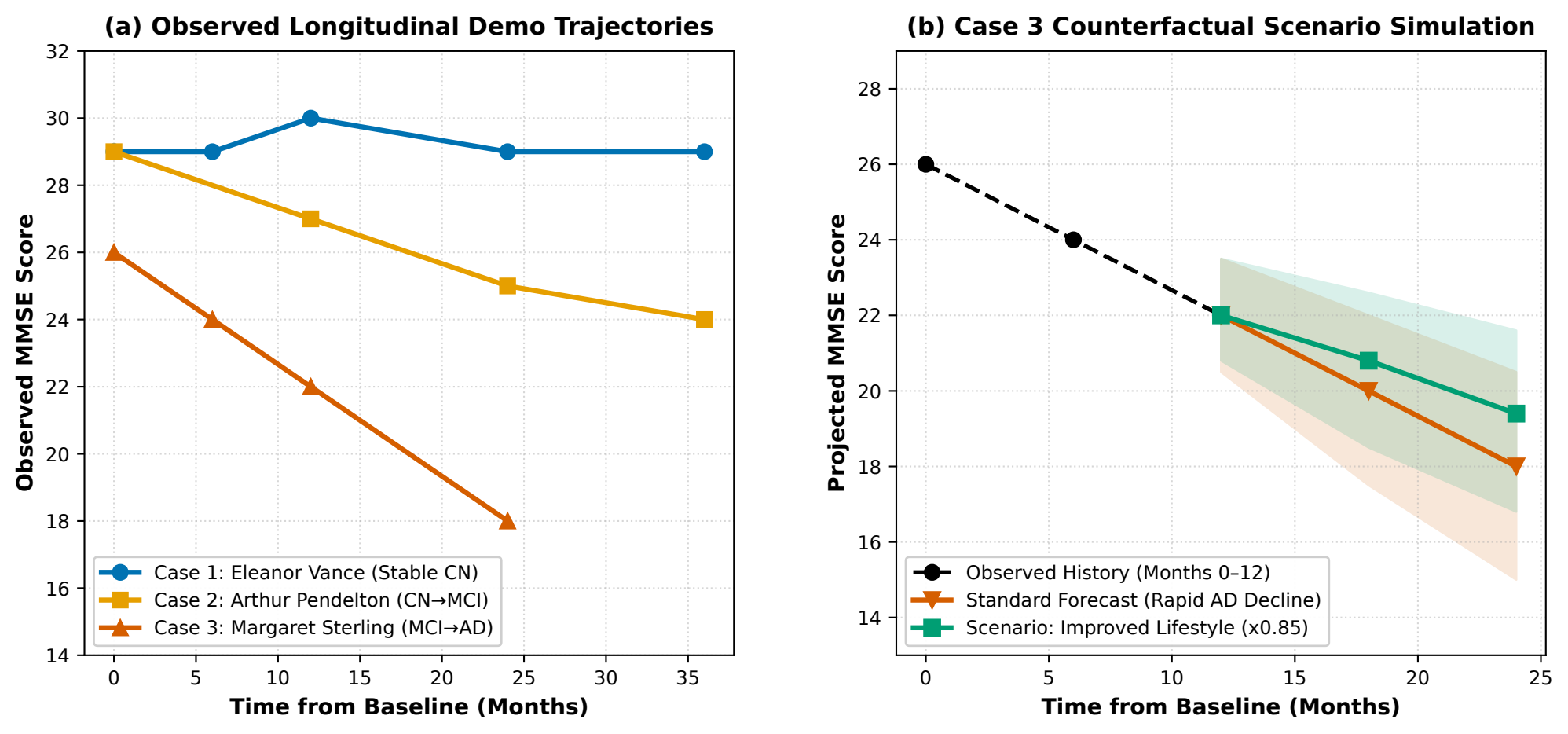
(a) Observed Longitudinal Demo Trajectories
Observed MMSE Score
Time from Baseline (Months)
Case 1: Eleanor Vance (Stable CN)
Case 2: Arthur Pendelton (CN→MCI)
Case 3: Margaret Sterling (MCI→AD)
(b) Case 3 Counterfactual Scenario Simulation
Projected MMSE Score
Time from Baseline (Months)
Observed History (Months 0–12)
Standard Forecast (Rapid AD Decline)
Scenario: Improved Lifestyle (x0.85)